



Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Decision Trees

WEEK 2 · DAY 4

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

A model that asks questions

Ask

a question of one column

Split

the accounts in two

Stop

the hard part: today's lesson

Allowed names

model probability of default in this file

flagged at threshold t

the model's score

Never called

a credit decision or an approval

a lending policy or a lending rule

a risk grade for a real person

Real UCI data: 30,000 accounts, Taiwan, 2005. No person column is a feature. Test rows stay closed.

A tree as questions

Word	Meaning
Node	A group of training accounts
Root	The first node: every account
Split	Is this value at most t? Yes left
Leaf	A node with no question
Depth	Questions from root to deepest leaf

```
|--- PAY_0 <= 1.50  
|   |--- weights: [13455.00, 2660.00] class: 0  
|--- PAY_0 > 1.50  
|   |--- weights: [563.00, 1322.00] class: 1
```

18,000 training accounts. Left: codes -2, -1, 0, 1 (two undocumented).
Right: a delay of two months or more.

PAY_0 <= 1.5: how this file splits, in its training rows. Not a rule for lending.

Gini impurity, and a split by hand

$Gini = 1 - p^2 - (1 - p)^2$ p = share of defaults; 0 pure, 0.5 half and half

Group	Accounts	Defaults	Share	Gini
all training rows	18,000	3,982	0.2212	0.3446
left: PAY_0 ≤ 1.5	16,115	2,660	0.1651	0.2756
right: PAY_0 > 1.5	1,885	1,322	0.7013	0.4189

0.2906

weighted Gini of the two sides

0.0539

decrease: the largest of any question

Four counts by COUNTIFS. Excel does it too.

The default tree overfits

38

questions deep

2,750

leaves

6.5

accounts per leaf

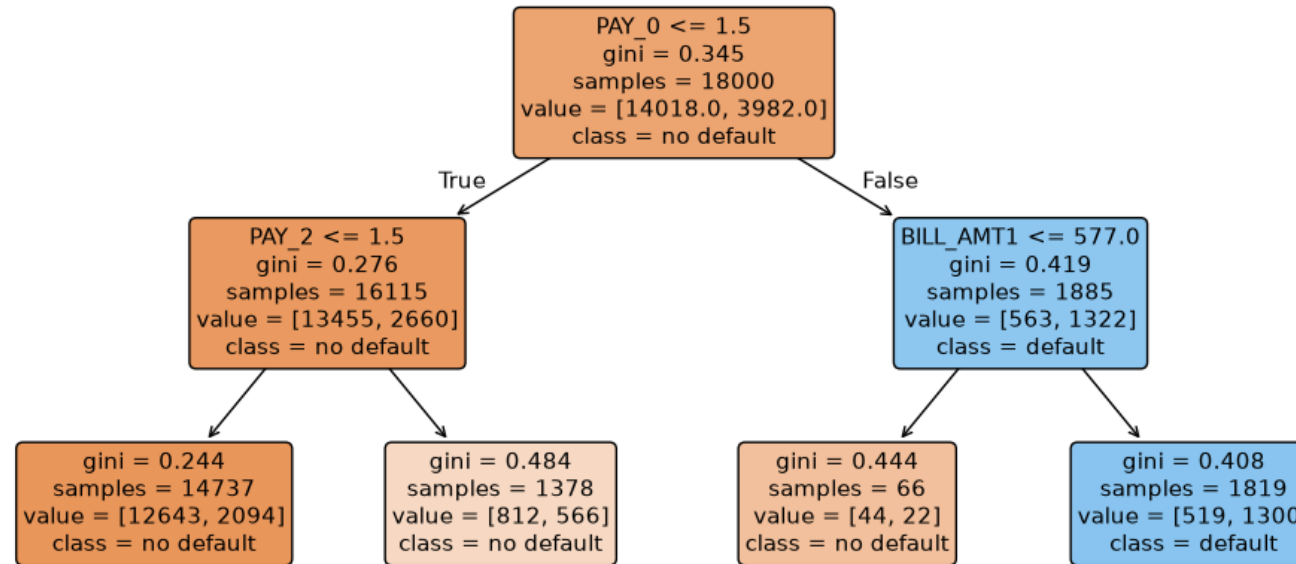
	ROC AUC	Recall at 0.5	Precision
Training rows	0.9997	0.9709	0.9895
Validation rows	0.6161	0.4032	0.3888

Training recall is not 1: 48 groups of identical accounts (413 in all) hold both outcomes.

Nearly memorised on training rows, much worse on rows it never saw: overfitting.

Reading a tree: depth 2

A tree of depth 2 on the 18,000 training accounts (UCI card file)



In this file's training rows, of accounts with status 2 or more and a bill above 577, 71.5 percent defaulted. A description, not a rule.

Class weights: make the rare class count

2.2602

weight of a default

0.6420

weight of a non-default

9,000

total weight of each class

Depth 3, validation rows	ROC AUC	Recall	Precision
Unweighted	0.7197	0.3662	0.6676
class_weight="balanced"	0.7491	0.5946	0.4524

Weighted output is a score: 0.2770 in a leaf of 8,346 accounts where 0.0981 defaulted.

Pre-pruning: stop the tree early

Setting	Stops
max_depth	a path longer than this
min_samples_leaf	a leaf smaller than this

```
cv=mlkit.cv_folds() # StratifiedKFold(5,  
                    #   shuffle=True, 42)
```

Chosen by five-fold cross-validation inside the 18,000 training rows. 38 tests pass.

depth 8
leaves of at least 200

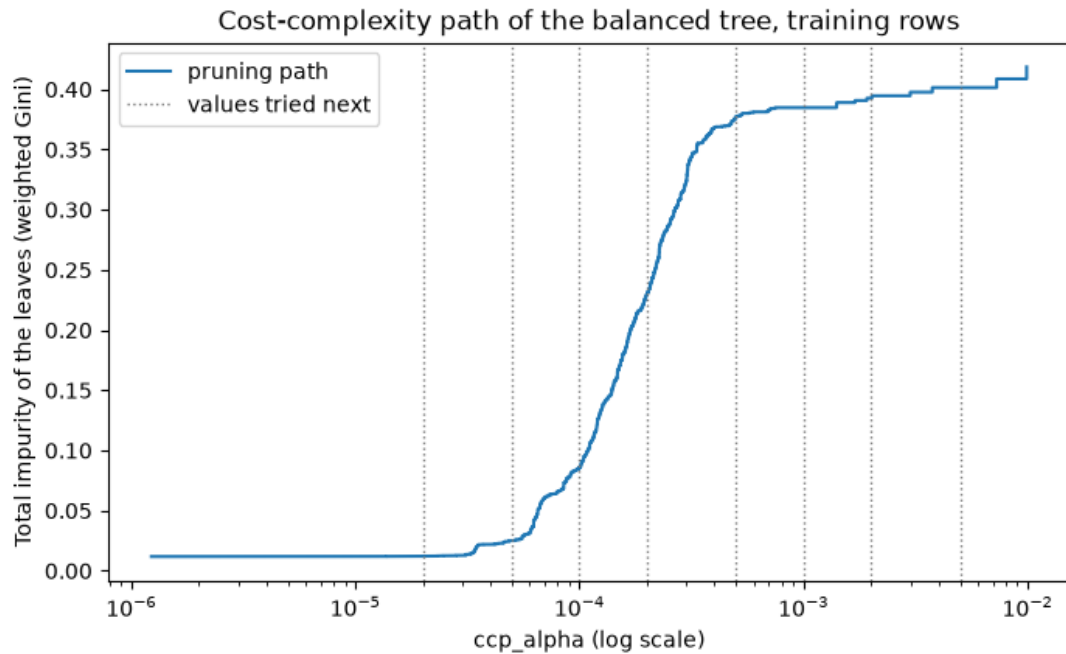
0.7664
cross-validated ROC AUC

52
leaves

0.7633
validation ROC AUC

The validation rows stay out of the choice, so they can report it.

Post-pruning: grow, then cut back



1,743 alphas; best ccp_alpha 0.001 (cross-validated 0.7598).

Validation rows	Depth	Leaves	ROC AUC
default	38	2,750	0.6161
default, balanced	49	3,233	0.6158
pre-pruned, balanced	8	52	0.7633
post-pruned, balanced	6	12	0.7627

Balanced trees ask $\text{PAY}_0 \leq 0.5$ first: weights change the questions.

Feature importance, and what it does not mean

Column	Impurity	Permutation
PAY_0	0.7115	0.1130
PAY_AMT2	0.0705	0.0108
LIMIT_BAL	0.0367	0.0185
BILL_AMT1	0.0303	0.0030
PAY_4	0.0293	0.0145

Not cause

The tree used it to sort accounts

Not direction

Higher value: more or fewer?

Shared

Related PAY_ columns mask each other

Training rows

Impurity favours many split points

Impurity: training rows. Permutation: fall in validation ROC AUC when a column is shuffled.

Scaling does not change a tree

A split asks above or below; scaling keeps the order of values and thresholds.

6,000

validation accounts, same output (pre-pruned tree)

5

accounts exactly on a threshold (default tree)

Same tree, node for node. A tree needs no scaling.

Excel side by side: a tree by hand

Excel side by side (tested in Excel)

`=COUNTIFS(A2:A18001,"<=1.5")`

`=COUNTIFS(A2:A18001,"<=1.5",D2:D18001,1)`

`=1-(G2/G1)^2-((G1-G2)/G1)^2`

`=G3/G1*G8+G5/G1*G9 =G7-G10`

`=AVERAGEIFS(D2:D18001,A2:A18001,">1.5",C2:C18001,">577")`

`=IF(A2<=1.5,IF(B2<=1.5,0,0),IF(C2<=577,0,1))`

`=H1/(H1+H2) =H1/(H1+H3)`

16,115 accounts on the left

2,660 defaults on the left

Gini 0.3446

0.2906, decrease 0.0539

leaf share 0.7147

the tree's flag, all rows

recall 0.3052, precision 0.6935

Excel has no decision tree. It can count, weigh, and apply a tree Python fitted.

Save the day: tag the pruned trees

1. Code

mlkit.py and test_mlkit.py

2. Unpruned

results.csv: two trees

3. Pruned

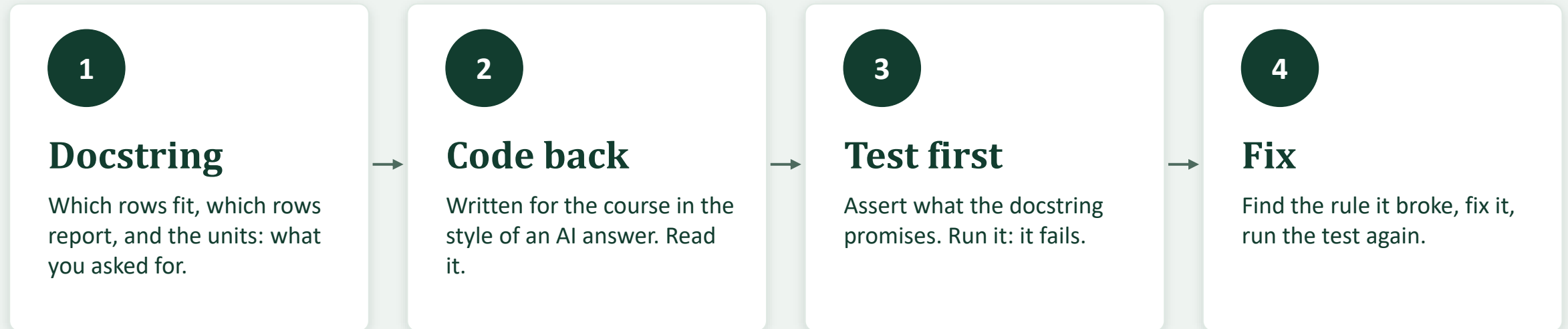
two more rows, then git tag tree-pruned

```
model,features,train_rows,valid_rows,depth,leaves,valid_roc_auc,valid_recall,valid_precision
default,19,18000,6000,38,2750,0.6161,0.4032,0.3888
"default, balanced",19,18000,6000,49,3233,0.6158,0.4160,0.3887
"pre-pruned, balanced",19,18000,6000,8,52,0.7633,0.6149,0.4682
"post-pruned, balanced",19,18000,6000,6,12,0.7627,0.5840,0.4805
```

A tag names one commit for good: git show tree-pruned:results.csv.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.



Today: the search is fitted on training and validation rows (24,000), then scored on validation: 0.7645.

Code that runs is not the same as code that is right.

What you can do now

- A tree picks the question with the largest Gini decrease
- Left to grow it memorises: 0.9997 training, 0.6161 validation
- Class weights make defaults count; the output becomes a score
- Prune before or after, chosen by cross-validation on training rows
- Importance says what the tree used, never why

Week 2, Day 5 Case study: the card file, a threshold for recall 0.70, test rows opened once